

1 Understanding word2Vec

(a) y is a hot-one vector so $y_w = \begin{cases} 1 & w = o \\ 0 & \text{otherwise} \end{cases}$

(b) (i) $\because J_{\text{naive-softmax}}(v_o, o, U) = -\log(\hat{y}_o) = -\log\left[\frac{\exp(\mu_o^T v_o)}{\sum_{w \in \text{Vocab}} \exp(\mu_w^T v_o)}\right]$ Let $M = \sum_{w \in \text{Vocab}} \exp(\mu_w^T v_o)$

$$\frac{\partial J}{\partial v_o} = \frac{\partial}{\partial v_o} (-\mu_o^T v_o) + \frac{\partial}{\partial v_o} \log(M) = -\mu_o + \frac{\sum_{w \in \text{Vocab}} \exp(\mu_w^T v_o) \cdot \mu_w}{M} = -\mu_o + \sum_{w \in \text{Vocab}} \hat{y}_w \cdot \mu_w$$

$$\text{use } U \rightarrow \frac{\partial J_{\text{naive-softmax}}(v_o, o, U)}{\partial v_o} = U(\hat{y} - y)$$

(ii) obviously when $\hat{y} = y$ the gradient is zero, means a perfect model: 100% (or 1) probability for the true outside word and 0 for all others.

(iii) subtracting $U\hat{y}$ decreases $\hat{y}_w$ which helps $\hat{y}_o$ for the true outside word; subtracting $-Uy$ increases $\mu_o^T v_o$ which also helps improving predicting for the correct word.

(c) for some $d > 1$ like $\mu_{\text{huge}} = d\mu_{\text{big}}$, L2 normalization make a $\frac{\mu_{\text{huge}}}{\|\mu_{\text{huge}}\|_2} = \frac{\mu_{\text{big}}}{\|\mu_{\text{big}}\|_2}$ which will remove useful difference

But for words without the concept of size, strength, etc, L2 could be useful.

(d) Case 1 $w = o$ $\frac{\partial J}{\partial \mu_o} = \frac{\partial}{\partial \mu_o} [\log(\hat{y}_o)] = -\frac{1}{\hat{y}_o} \cdot \frac{\partial \hat{y}_o}{\partial \mu_o} = -\frac{1}{\hat{y}_o} \cdot \frac{\partial}{\partial \mu_o} \left(\frac{e^{\mu_o^T v_o}}{M} \right) = (\hat{y}_o - y_o) v_o$

(e) Case 2 $w \neq o$ $\frac{\partial J}{\partial \mu_w} = -\frac{1}{\hat{y}_o} \cdot \frac{\partial \hat{y}_o}{\partial \mu_w} = (\hat{y}_w - y_w) v_o$ use U definition and combine

$$\therefore \frac{\partial J_{\text{naive-softmax}}(v_o, o, U)}{\partial U} = (\hat{y} - y)^T v_o \text{ for all } w \in \text{Vocab}$$

~~for~~ (f) obviously $f'(x) = \begin{cases} 1 & , x > 0 \\ \alpha & , x < 0 \end{cases}$

(g) $\sigma'(x) = \frac{e^x(e^{x+1} - e^x \cdot e^x)}{(e^{x+1})^2} = \frac{e^x}{(e^{x+1})^2} = \frac{(e^{x+1})^2}{(e^{x+1})^2} = \frac{e^x}{e^{x+1}} \cdot (1 - \frac{e^x}{e^{x+1}}) = \sigma(x)(1 - \sigma(x))$

(h) (i) $\frac{\partial J}{\partial v_o} = -\frac{1}{\sigma(\mu_o^T v_o)} \cdot \sigma(\mu_o^T v_o) (1 - \sigma(\mu_o^T v_o)) \cdot \mu_o + \sum_{s=1}^K \frac{1}{\sigma(\mu_{w_s}^T v_o)} \cdot \sigma(-\mu_{w_s}^T v_o) (1 - \sigma(-\mu_{w_s}^T v_o)) \cdot \mu_{w_s}$
 ① $= (\sigma(\mu_o^T v_o) - 1) \mu_o + \sum_{s=1}^K \sigma(\mu_{w_s}^T v_o) \mu_{w_s}$

(h) (i) ② $\frac{\partial J}{\partial \mu_0} = (\sigma(\mu_0^T V_c) - 1) V_c$ the other part does not depend on μ_0

③ $\frac{\partial J}{\partial \mu_{w_s}} = \sigma(\mu_{w_s}^T V_c) V_c$

(ii) reuse V_c $\frac{\partial J}{\partial V_c} = \bigcup_{i \in \{w_1, \dots, w_k\}} \left[(\sigma(\mu_0^T V_c) - 1) \sigma(\mu_{w_1}^T V_c) \dots \sigma(\mu_{w_k}^T V_c) \right]^T$

(iii) the efficiency comes from reducing the number of words required (only $k+1$)

(i) now we should consider how many sampled words are equal to w_s in $\{w_1, \dots, w_k\}$

Let t_s be equal to the times word w_s appeared in $\{w_1, \dots, w_k\}$

$$\frac{\partial J}{\partial \mu_{w_s}} = -\log(\sigma(\mu_0^T V_c)) - t_s \log(\sigma(\mu_{w_s}^T V_c)) - (k - t_s) \log(\sigma(\mu_{w_s}^T V_c))$$

there's only one part we need $\frac{\partial J}{\partial \mu_{w_s}} = t_s \cdot \sigma(\mu_{w_s}^T V_c) \cdot V_c$ where t_s means times of w_s

(j) (i) $\frac{\partial J_{\text{skip-gram}}(V_c, w_{t-m}, \dots, w_{t+m}, U)}{\partial U} = \sum_{\substack{-m \leq j \leq m \\ j \neq 0}} \frac{\partial J_{\text{skip-gram}}(V_c, w_{t+j}, U)}{\partial U}$

(ii) $\frac{\partial J_{\text{skip-gram}}(V_c, w_{t-m}, \dots, w_{t+m}, U)}{\partial V_c} = \sum_{\substack{-m \leq j \leq m \\ j \neq 0}} \frac{\partial J_{\text{skip-gram}}(V_c, w_{t+j}, U)}{\partial V_c}$

(iii) $\frac{\partial J_{\text{skip-gram}}(V_c, w_{t-m}, \dots, w_{t+m}, U)}{\partial V_w} = 0$ when $w \neq c$